

Some review questions - 6th week

Q1 Give the solution set of the system of equations:

$$\begin{cases} x_1 + 3x_3 - 2x_4 = 0, \\ x_2 + x_3 - 3x_4 = 0, \\ 0 = 1. \end{cases}$$

Q2 Give the solution set of the system of equations:

$$\begin{cases} x_1 + x_2 = 3, \\ x_3 + x_4 = -4, \\ x_5 = 2. \end{cases}$$

Q3 Is the empty set $\emptyset$ a linear space? Is the set $\{\mathbf{0}\}$ a linear space, where $\mathbf{0}$ is the size-n zero vector?

Q4 For each of the following sets tell if they are linear spaces or not. If the answer is no, explain why.

- $\mathbb{R}^{2 \times 3}$
- $\{0, 1, 2, \dots\}$
- $\{(-2, 1), (0, 0), (2, -1)\}$
- $\mathbb{R} \cup \mathbb{R}^2$.

Q5 Give the definition of a subspace (define clearly the setting).

Q6 Provide criteria to check that a subset W of a linear space V is a subspace.

Q7 Is $\{(0, \dots, 0)\}$ a subspace of $\mathbb{R}^n$? Is $\mathbb{R}^n$ a subspace of $\mathbb{R}^n$? In both cases, prove your answer.

Q8 Is the set of upper triangular matrices of size $n \times n$ a linear space? Prove your answer.

Q9 Is the set of elementary matrices of size $n \times n$ a linear space? Prove your answer.

Q10 Give the definition of linear dependence (for k elements in a linear space $u_1, \dots, u_k$).

Q11 Give the definition of linear independence (for k elements in a linear space $u_1, \dots, u_k$).

Q12 Are the columns of the matrix $\begin{bmatrix} 1 & 0 & a & 0 & c \\ 0 & 1 & b & 0 & d \\ 0 & 0 & 0 & 1 & e \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ linearly dependent or linearly independent? Prove your result.

Q13 Are the vectors $(1, 1)$ and $(2, 2)$ linearly dependent? Justify. Represent them on axes.

Q14 Are the vectors $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$ linearly dependent? Justify.

Q15 Given two elements $u, v \in V$ of a linear space V , are the elements $u, v, u + v$ linearly independent? Justify.

Q16 Represent graphically in $\mathbb{R}^2$ two linearly independent vectors and in a second graph two linearly dependent vectors.

Q17 What can be said of the solutions of the equation $Ax = 0$ when A has dependent columns? Prove your result.

Q18 Express the intersection $S \cap T$ of $S = \{x \in \mathbb{R}^3 : 3x_1 + x_2 - x_3 = 0\}$ and $T = \{x \in \mathbb{R}^3 : x_1 - x_2 = 0\}$. Are S, T and $S \cap T$ linear spaces (no need for justification).

Q19 Give the definition of a homogeneous system.

Q20 Prove that the solution set of a homogeneous system $Ax = 0$ is a linear space.

Q21 Give the definition of the null space of a matrix $A \in \mathbb{R}^{m \times n}$.

Q22 Give the definition of the span of a set of n elements of a linear space $u_1, \dots, u_n$.